A Covariant Action for the Elastic Dark-Energy Medium

The Branch-B field theory written: a relativistic two-invariant elastic solid whose volumetric sector carries the $a_0 = c^2\sqrt{\Lambda/32\pi}$ displacement law and terminates at the derived entropy-budget cutoff $y_c = Z/2$ — turning the posited shear fraction w and the Cassini quadrupole into defined computations

Carl P. Zimmerman · Briar Creek Tech · 2026-07-11 (v3 — the two “defined computations” of §3 are now *carried out*: the Cassini quadrupole reduces to a single free material scalar, the shear Poisson ratio, which the action does not fix and whose every natural value marginally fails; v2 corrected v1′s sourcing claim to the nonminimal $[1 + h(J)]R$)

Abstract

The elastic-medium branch (Branch B) of the de Sitter–Unruh a_0 program has until now been a constitutive prescription without an action — its structural debt (P1). This note writes the action. The medium is a relativistic elastic solid in the standard three-scalar formulation [Carter & Quintana 1972; Bucher & Spergel 1999; Battye & Moss 2007; Endlich, Nicolis & Wang 2013]: fields $\varphi^a(x)$, $a = 1, 2, 3$, internal metric $B^{ab} = g^{\mu\nu}\partial_\mu\varphi^a\partial_\nu\varphi^b$, with the medium's rest frame, number density, volumetric strain J, and deviatoric (shear) strain Σ all built from B^{ab}. The Lagrangian is

$$S = \int\sqrt{-g}\left[\frac{c^4\,[1+h(J)]\,R}{16\pi G} - \rho_\Lambda c^2\mathcal{E}(J,\Sigma)\right] + S_{\rm baryons}[g,\psi], \qquad \mathcal{E} = 1 + \frac{V(J) + \mu_s\,\Sigma}{\rho_\Lambda c^2},$$

where the relaxed medium *is* the dark energy ($\mathcal{E}(0,0) = 1$), and the constitutive content — all of it inherited from previously machine-verified results, none newly fitted — is: **(i)** a volumetric potential $V(J)$ whose deep-regime branch is the scale-free law $V'(J) \propto \sqrt{J}$ with total deep stiffness $K_{\rm eff} = a_0^2/16\pi G$, fixed by matching the Verlinde displacement response $g_D = \sqrt{a_{0V}\,g_{\rm bar}/6}$ at $a_{0V} = cH_\Lambda = Za_0$ (the match that lands at $\sqrt{Z/6} = 0.982$ of the required lensing source); **(ii)** a domain boundary at $\mid J \mid = J_c$ implementing the entropy-budget constraint $S_M \le S_{DE}$, whose location is the derived cutoff $y_c = Z/2$ (pinned to Verlinde's published coefficients) — the budget lives outside the constitutive law, exactly as the single-valued-F impossibility theorem requires; **(iii)** a linear shear sector with modulus μ_s, bounded above by the exact refusal $\rho_\Lambda c^2/K_{\rm eff} = 6Z^2 = 201.1$ (the medium's shear waves are material sound at $v_T = \sqrt{\mu_s/\rho_\Lambda} \le 0.17c$, not gravitational waves — metric tensor modes propagate at c untouched, so GW170817 is satisfied in every sector by construction, and light bends by the medium's genuine stress-energy in the one shared metric). **Corrected in v2 (a self-caught overclaim):** the *minimally coupled* solid cannot gravitationally source the halo — its strain energy falls short of the required apparent-dark-matter density by $10^{6.5-7}$ (computed; the v1 script never actually checked this item). The mechanism is therefore **nonminimal**, as in Verlinde's own construction: the volumetric strain modulates the gravitational response through $[1 + h(J)]R$, with h fixed at the weak-field level by $\nabla^2 h = (8\pi G/c^2)\rho_D$ — giving $h \sim 10^{-6}$ everywhere (weak), deep-Newton-suppressed at high accelerations ($\nabla h \to 2(\nu - 1)g/c^2 \to a_0/c^2$), and coupled to the *trace* sector only, so the two-invariant monopole/quadrupole decoupling and the shear-fraction analysis are preserved

unchanged. Machine-verified consequences of the corrected action: the nonminimal weak-field limit reproduces the required apparent-dark-matter source (now checked, not asserted); the background cosmology returns $\rho_{\mathrm{DE}} = \rho_\Lambda$ at the relaxed point with $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\mathrm{DE}}(z)}$ inherited automatically (the falsifiable $a_0(z) \propto \sqrt{\rho_{\mathrm{DE}}}$ evolution is now a property of the action, not a separate posit); the phonon spectrum is ghost-free and stable for $V'' > 0$, $\mu_s > 0$; and the graviton mass induced by the shear sector is 10^{-42} eV-scale, negligible. What the action *newly defines rather than solves*: the shear fraction w and the medium's Cassini quadrupole Q_2 are boundary-value computations of the written theory. **Carried out in v3** (two independent solvers, §3): both reduce to $Q_2 = Q_2^{\mathrm{scalar}}/(1 + 4\beta/\kappa_t)$, with $\kappa_t = V''(1)/K_{\mathrm{eff}} = 0.5$ *pinned* (footing-independent) and $\beta = \mu_s/3K_{\mathrm{eff}}$ the shear share; Cassini passes iff $\beta > \beta_{\mathrm{crit}} \approx 0.40$. The action fixes only the combination $K_F + \mu_s/6 = K_{\mathrm{eff}}$, leaving β — the **shear Poisson ratio** — a genuinely free material parameter (bounded only to $(0, 2)$ by ghost-freedom and causality); no microphysical principle in the action pins it, and cosmological rigidity bounds on a $w = -1$ solid do not cap it below β_{crit}. So Cassini is decided by that one scalar, and every *natural* elasticity value ($\nu = 1/4 \Rightarrow \beta = 2/7$; the Verlinde magnitude $\Rightarrow \beta = 1/3$) sits below β_{crit} and **marginally fails** ($\times$ 1.1–1.8): the branch is formally open but the weight of evidence tilts to fail. The MOND sign remains an input; a_0's value and Z remain inputs; this is Branch B *written*, not adopted, and not a theory of everything.

1. The construction

Kinematics (standard relativistic elasticity). Three scalars $\varphi^a(x)$ label the medium's elements; the internal metric is $B^{ab} = g^{\mu\nu}\partial_\mu\varphi^a\partial_\nu\varphi^b$. The number density is $n = n_0\sqrt{\det B^{ab}}$ (with n_0 the relaxed density), the medium four-velocity u^μ is the unique timelike direction with $u^\mu\partial_\mu\varphi^a = 0$, the **volumetric strain** is $J = -\frac{1}{2}\ln(\det B/\det\bar{B})$ relative to the relaxed configuration $\bar{B}$, and the **shear invariant** is $\Sigma = \left[\frac{1}{2}\operatorname{tr}\tilde{B} - 3\right]$ built from the unimodular part $\tilde{B}^{ab} = B^{ab}/(\det B)^{1/3}$ (for small strains, $\Sigma \to \frac{1}{2}e_{ij}e^{ij}$, the squared deviatoric strain — the J_2 of the campaign lemmas). Internal $SO(3)$ symmetry among the φ^a enforces isotropy; the two invariants (J, Σ) are the *only* independent scalars to this order — the two-invariant structure that decouples the monopole (bulk, $\ell = 0$) from the quadrupole (shear, $\ell = 2$), which the campaign proved scalar AQUAL cannot do.

Dynamics. $S_{\mathrm{med}} = -\int\sqrt{-g}\rho_\Lambda c^2\,\mathcal{E}(J,\Sigma)$ with $\mathcal{E} = 1 + [V(J) + \mu_s\Sigma]/\rho_\Lambda c^2$ and:

- **The relaxed point is the dark energy.** $V(0) = 0$, $\Sigma = 0$: the medium contributes $T_{\mu\nu} = -\rho_\Lambda c^2 g_{\mu\nu}$ — an equation-of-state $w_{\mathrm{eos}} = -1$ fluid. The medium does not *add* a dark sector; it *is* the dark energy, deformed.
- **The volumetric sector carries the MOND response.** For $0 < J < J_c$ the potential has the scale-free branch $V'(J) = 2K_{\mathrm{eff}}\sqrt{J/\kappa^2}$-normalized such that the static response to a baryonic source reproduces the displacement law $g_D = \sqrt{a_{0V}g_{\mathrm{bar}}/6}$, with the total deep stiffness (volumetric plus the shear share fixed by the compatibility lemma $J_2 = J_1^2/6$) equal to $K_{\mathrm{eff}} = a_0^2/16\pi G = 2.6 \times 10^{-12}$ Pa. This is the $\sqrt{Z/6} = 0.982$ match, now a property of V.
- **The budget boundary.** The domain of V ends at J_c (equivalently the response terminates at $y_c = Z/2$): implemented as the constrained sector $J \le J_c$ with multiplier — *not* as a feature of V itself, honoring the impossibility theorem that no single-valued constitutive law can carry the tail. Beyond saturation the medium's displaced entropy budget is exhausted and its response depletes as derived ($T = \min(1, y_c/y)$, the reading supported by Verlinde's text, with the cap alternative computed dead).

- **The shear sector is near-linear by construction and bounded by theorem.** $\mu_s \Sigma$ with constant μ_s: transverse phonons at $v_T = \sqrt{\mu_s/\rho_\Lambda}$. The exact refusal $\rho_\Lambda c^2/K_{\rm eff} = 6Z^2$ caps $v_T \le 0.17c$ if the deep amplitude is to survive — the medium's shear waves are slow material sound. The *anharmonic* shear fraction w (the residual coupling of shear to the volumetric nonlinearity through det-geometry) is no longer a free parameter of a prescription: it is a **computable functional of this action**, $w[V, \mu_s]$, via the second-order expansion around the galactic pre-strain state — the campaign's band $w \sim [0.08, 2.7]$ is the pre-action estimate; the action defines the exact object.

Couplings. Baryons couple to $g_{\mu\nu}$ only; photons couple to $g_{\mu\nu}$ only; the medium couples to $g_{\mu\nu}$ only. One metric. Light bends because the deformed medium's $T_{\mu\nu}$ *gravitates* — a genuine source, no disformal structure, no second cone: GW170817 is satisfied identically ($c_\gamma = c_{\rm GW} = c$), which is the constraint that created Branch B in the first place.

2. Machine-verified consequences (committed script)

1. **Static weak field (corrected in v2).** The medium's own stress-energy is $\sim 10^{6.5-7}$ too small to source the halo (audited: $K_{\rm eff}\varepsilon^2/c^2 \lesssim 3 \times 10^{-29}$ kg m^{-3} vs the required $\sim 3 \times 10^{-22}$ at 10 kpc) — v1 asserted this item without checking it, reproducing the program's earlier bath-energetics kill at the medium's own strain energy. The corrected sourcing is the **nonminimal sector**: linearizing $[1 + h(J)]R$ gives $\nabla^2\Phi = 4\pi G\rho_b + (c^2/2)\nabla^2 h$, so $\rho_D = (c^2/8\pi G)\nabla^2 h$; matching fixes $h = (2/c^2) \times$ the MOND excess potential, i.e. $h \sim 10^{-6}$ (a weak dressing of GR), with the deep-branch closed form $h \propto (\sqrt{a_0 GM}/c^2) \ln r$ and $h(J)$ obtained by eliminating r on the matched background. High-g: $\nabla h \propto (\nu - 1)g \to a_0/2$ — the same deep-Newton suppression the modified-inertia evasion used, so the h-channel monopole is solar-system-safe.
2. **Cosmology and $a_0(z)$.** On FRW the relaxed medium gives $\rho_{\rm DE} = \rho_\Lambda$, $w_{\rm eos} = -1$; promoting the relaxed density to the measured dark-energy density, the deep stiffness scales as $K_{\rm eff}(z) = a_0(z)^2/16\pi G$ with $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\rm DE}(z)}$ — the program's one distinctive falsifiable prediction now follows from the action's own normalization instead of being separately postulated. If dark energy is a true cosmological constant, a_0 is constant and the distinctive content dissolves, exactly as pre-registered.
3. **Degrees of freedom and stability.** Three phonons (one longitudinal, $c_L^2 = (V'' + \frac{4}{3}\mu_s)/\rho_\Lambda$; two transverse, $c_T^2 = \mu_s/\rho_\Lambda$), ghost-free and gradient-stable for $V'' > 0, \mu_s > 0$; the metric tensor sector is standard GR (two polarizations at c); the shear-induced graviton mass $m_{\rm GW}^2 \sim 16\pi G\mu_s/c^4$ is of order the Hubble scale squared — utterly negligible against all bounds.
4. **The exact numbers carried.** $K_{\rm eff} = a_0^2/16\pi G$, $\rho_\Lambda c^2/K_{\rm eff} = 6Z^2 = 201.06$, $v_T^{\max} = c\sqrt{6}/(6Z^2)^{1/2}$-normalized $\le 0.17c$, $y_c = Z/2 = 2.894$ — all reproduced symbolically from the action's own definitions, both footings.

3. The Cassini computation carried out — it reduces to one free scalar

The v1/v2 promise was that Branch B's Cassini quadrupole is a computation, not a posit. This section carries it out and reports where it bottoms.

The reduction (two independent solvers). The medium's Q_2 around the Sun in the galactic external field was solved two ways: an $\ell = 2$ Navier radial boundary-value problem (spheroidal Takeuchi–Saito variables), and a 2D axisymmetric variational energy minimization. Both were gated on reproducing the single-invariant (QUMOND) scalar quadrupole in the $\mu_s \to 0$ limit, and both do ($Q_2^{\rm scalar} = 2.0$–

$2.5 \times 10^{-26}\,\mathrm{s}^{-2}$ canonical, with the correct external-field trend). They then reduce to the **identical P-wave form**

$$Q_2 = w\, Q_2^{\mathrm{scalar}}, \qquad w = \frac{K_t}{K_t + \frac{4}{3}\mu_s} = \frac{1}{1 + 4\beta/\kappa_t},$$

where $\kappa_t = K_t/K_{\mathrm{eff}}$ is the bulk *tangent* stiffness at the quadrupole-sourcing shell ($r \sim r_t \sim 4700$ AU) and $\beta = \mu_s/3K_{\mathrm{eff}}$. The physics this settles: shear rigidity genuinely suppresses the $\ell = 2$ of the volumetric strain ($w < 1$) — the "shear relaxes for free via divergence-free displacement" escape is geometrically excluded, since only the $\ell = 0$ mode is shear-free.

One scalar is pinned, one is free. κ_t follows from the written $V'(J) = 2K_{\mathrm{eff}}\sqrt{J/\kappa^2}$ as the exact identity $\kappa_t = V''(1)/K_{\mathrm{eff}} = 1/\kappa = 0.5$ (footing-independent; the secant-vs-tangent ambiguity is decided — not chosen — by the displacement match anchoring a first-derivative/stress response, giving $V''(1) = \frac{1}{2}K_{\mathrm{eff}}$). Against the ceiling $Q_2 < 5.2 \times 10^{-27}$, this makes Cassini pass iff $\beta > \beta_{\mathrm{crit}} = \kappa_t(Q_2^{\mathrm{scalar}}/5.2 \times 10^{-27} - 1)/4 \approx 0.40$ (canonical) / 0.60 (alt). But β is **not** pinned: the action writes only the single relation $K_F + \mu_s/6 = K_{\mathrm{eff}}$ (the $\ell = 0$ Verlinde monopole combination), fixing the *sum* while leaving the shear/bulk split — the shear **Poisson ratio** — free. This is a genuine feature of isotropic elasticity (two independent moduli), not an omission: the three-scalar solid EFT [Endlich–Nicolis–Wang 2013] has shear $\sim (F_Y + F_Z)$ and bulk $\sim F_{XX}$ as independent Lagrangian data, and $w_{\mathrm{eos}} = -1$ with the internal $SO(3)$ fixes only the *form*.

Neither microphysics nor cosmology closes it. Verlinde's own emergent elasticity does pin a Poisson ratio, but a framework-*inadmissible* one (Poisson $\nu = +1$, vanishing P-wave modulus, negative bulk) excluded by the theory's own ghost wall $K_F > 0$. On the cosmological side the medium is a $w = -1$ *solid* whose shear sources anisotropic stress; the finite rigidity variable is $\mu_s/(\rho_\Lambda c^2) = \beta/(2Z^2) = \beta/67$ (the customary $c_v^2 = \mu/(\rho + P)$ diverges at $w = -1$ and must not be used), and the induced late-time gravitational slip $\sim \Omega_{\mathrm{DE}}\,\beta/67 \sim 0.4$–$2\%$ sits well below current ($\sim 10\%$) sensitivity — cosmology permits the entire mechanical window and does **not** cap β below β_{crit}. (The tight published shear bound $\mu/\rho \lesssim 10^{-5}$ is a *radiation-like* solid at recombination — wrong species and epoch; transferring it would manufacture a spurious failure.)

Verdict. Cassini is decided by a single genuinely free material scalar, $\beta \in (0, 2)$, which straddles β_{crit}. It is not a coin flip: every natural closure — Poisson-$\frac{1}{4}$ ($\beta = 2/7 = 0.286$), the Verlinde shear magnitude ($\beta = 1/3$), the natural-elasticity cluster $[0.18, 0.33]$ — lands *below* β_{crit} and marginally fails ($\times$ 1.1–1.8 over the ceiling; the friendliest admissible corner sits essentially *on* it). A pass exists only for a medium shear-stiffer than any natural elasticity motivates ($\beta \gtrsim 0.40$, $\mu_s \gtrsim 1.25\,K_{\mathrm{eff}}$). So the branch is **formally open but evidence-tilted to fail**: the action has made Cassini falsifiable-by-one-number, and that number's natural values do not pass. This is the honest floor of the Branch-B quadrupole question — not the clean pass v1 hoped for, nor a clean kill.

4. Honest ledger

New here: the covariant action itself — relativistic elastic *linear* dark energy is prior art [Bucher & Spergel 1999; Battye & Moss 2007], and the solid-EFT kinematics are standard [Endlich–Nicolis–Wang 2013]; what has not previously been written is the **Verlinde-class anharmonic volumetric sector** (the scale-free $\sqrt{J}$ branch matched to $a_0 = c^2\sqrt{\Lambda/32\pi}$) **with the entropy-budget domain boundary at the derived $\boldsymbol{y_c = Z/2}$** — the constitutive content that makes an elastic solid produce

MOND-scale phenomenology. **Inherited, not new:** every matching coefficient (0.982, $y_c = Z/2$, $6Z^2$, K_{eff}) comes from the previously published, machine-verified campaign results — nothing is re-fitted here. **Still inputs:** the MOND sign, a_0's value, Z; and (new, honest) the *form* of $h(J)$ is matched to the displacement law, not derived — the constitutive debt moves from the (refuted) minimal sourcing claim into the nonminimal coupling, and the theory is now explicitly scalar-tensor-like: Branch B is modified gravity with a medium, as the program's ledger has always stated. **The one free parameter that decides Cassini (v3):** the shear Poisson ratio β — provably not fixed by the action, not pinned by microphysics, not capped by cosmology below the deciding value; every natural value marginally fails. Closing Cassini outright now requires an *independent* determination of that single modulus, which the present theory does not supply. **Still open beyond it:** the medium's full nonlinear well-posedness (the constrained sector's dynamics at the budget boundary); cluster-scale performance; and the Branch A/B identity choice — *writing* Branch B's action is not *adopting* it, and the observational fingerprint distinguishing it from Branch A remains indistinguishable at current sensitivity. No completeness or theory-of-everything claim is made.

Reproducibility

`real_research/reviews/shear_linearity_2026/elastic_action_consequences.py` (items 2–6), `action_source_audit_and_fix.py` (the v2 correction: the minimal-sourcing audit and the nonminimal weak-field derivation), `action_w_q2_computation.py` (the pinned κ, reconstructed (S, f_1)), and the v3 Cassini campaign in `../branchB_q2_gate_2026/vector_elastic_w_2026/` (the two-solver reduction `methodA_ode.py`/`methodB_fem.py`, the $\kappa_t = 0.5$ and β derivations `lane1_kappat.py`/`lane2_beta.py`, and the Poisson-ratio closure attempts `routeA_cosmo.py`/`routeB_micro.py`) plus the campaign scripts previously committed (`p6_pin_verlinde_coefficients.py`, `p6_throttle_fingerprint_sparc.py`, `lane1_theorem.py`, `../branchB_q2_gate_2026/`, `../lensing_source_fork_2026/`). Program concept DOI: 10.5281/zenodo.21253644; the $y_c = Z/2$ note: 10.5281/zenodo.21300855.

References

1. B. Carter & H. Quintana, Proc. R. Soc. A **331**, 57 (1972). 2. M. Bucher & D. Spergel, Phys. Rev. D **60**, 043505 (1999). 3. R. Battye & A. Moss, Phys. Rev. D **76**, 023005 (2007). 4. S. Endlich, A. Nicolis & J. Wang, JCAP **10**, 011 (2013). 5. E. Verlinde, SciPost Phys. **2**, 016 (2017). 6. M. Milgrom, ApJ **270**, 365 (1983). 7. S. Boran et al., Phys. Rev. D **97**, 041501 (2018). 8. H. Desmond, A. Hees & B. Famaey, MNRAS **530**, 1781 (2024).

Both a_0 footings; every coefficient inherited from committed, verified computations; Branch B written, not adopted.